INCH-POUND

MIL-G-53082(ME)
14 August 1989

MILITARY SPECIFICATION

GENERATOR SETS, DIESEL ENGINE DRIVEN

3kW, 60 HERTZ, 400 HERTZ, AND 28 VOLT DIRECT CURRENT

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the military design, tactical utility, air-cooled, diesel engine driven (DED) generator sets (herein after referred to as sets or set), 3 kilowatts (kW), 60 hertz (Hz) and 400Hz and 28 volt direct current (Vdc) and an optional acoustic suppression kit.

1.2 Classification.

1.2.1 Mode and size. The set shall be type I (tactical), class 2 (utility), size 3 (3kW) and of the following modes as specified (see 6.2):

Mode II - 400Hz
Mode III - 60Hz
Mode IV - 28 Vdc

1.2.2 Voltage connections and set rating. Mode IV generator sets shall be rated 28 Vdc. The mode II and mode III generator sets shall be rated 0.8 power factor (pf) lagging and shall be reconnectable for the following voltage ratings:

a. 1 phase, 2 wire, 120 volts.
b. 1 phase, 3 wire, 120/240 volts.
c. 3 phase, 3 wire, 120 volts line-to-line.
d. 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6115
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-R-550	- Rod, Ground (with attachments).
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-14105	- Paint, Heat-Resisting (For Steel Surfaces).
MIL-F-16884	- Fuel, Naval Distillate.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.
MIL-F-46162	- Fuel Diesel Referee Grade.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Artic.
MIL-C-53072	- Connector, Electrical, Power.
MIL-G-53082/1	- Generator Sets, Diesel Engine Driven 3kW, 400 Hertz.
MIL-G-53082/2	- Generator Sets, Diesel Engine Driven 3kW, 60 Hertz.
MIL-G-53082/3	- Generator Sets, Diesel Engine Driven 3kW, 28 Volt, Direct Current.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-705	- Generator Sets, Engine Driven, Methods of Tests and Instructions.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-1474	- Noise Limits for Army Material.
MIL-STD-889	- Dissimilar Metals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

HANDBOOKS

MILITARY

MIL-HDBK-705 - Generator Sets, Electrical Measurements and Instrumentation Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

TA 84-016 - Generator Set Assembly 3kW, 60 Hertz.
TA 84-021 - Generator Set Assembly, 3kW, 400 Hertz.
TA 84-026 - Generator Set Assembly, 3kW, 28 VDC.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FG, Fort Belvoir, VA 22060-5606.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail requirements for individual sets. The individual set requirements shall be as specified herein and in the applicable specification sheets. In the event of any conflict among the requirements thereof, the order of precedence shall be the detail specification, the general specification, and the drawing.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the set. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.2 Material deterioration prevention and control. The generator sets shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

deterioration that may be encountered in any of the applicable operating and storage environments to which the generator sets may be exposed.

3.1.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Starting, operating, handling, and storage.

3.3.1 Starting and operating. When equipped with all accessories, the set shall start (see 6.4.1) and operate within its rating as follows:

3.3.1.1 Starting.

3.3.1.1.1 Without winterization system. Within five minutes, using set batteries, the set shall start as follows:

a. At ambient temperatures from +125 °F (51.7 °C) to -25 °F (-31.7 °C) at all possible relative humidity (rh).
b. With 4 (±1) inches of rain per hour impinging on the set at angles from the vertical up to 45 degrees from all directions.
c. With base of the set in planes from level to 15 degrees from level in any direction.
d. At any altitude up to and including 8,000 feet above sea level (564.9 millimeters of mercury (mm Hg)) at ambient temperature of 95 °F (35 °C).
e. At any altitude up to and including 5,000 feet above sea level (632.4 mmHg) at an ambient temperature of 107 °F (41.7 °C).

3.3.1.1.2 With winterization system. When the winterization system is used at ambient temperature from -25 °F (-31.7 °C) to -50 °F (-45.6 °C), the set shall start within five minutes after use of the fuel burning winterization system for a maximum of 55 minutes.

3.3.1.2 Operating. Immediately after starting, the sets shall run at no load for 15 minutes (maximum warm-up period) except no warm-up period is permitted for the requirements of 3.3.1.1.2. The sets shall operate at all loads, continuous or intermittent, up to and including rated load (see 6.4.19) under any of the conditions or combination of the conditions specified in 3.3.1.1, except for the altitude condition (see 3.3.1.1.1d.) the set shall meet the above conditions at rated load for 5,000 feet (632.4 mmHg) at 107 °F (41.7 °C), and 90 percent of rated load at 8,000 feet (564.9 mmHg) at 95 °F (35 °C). Maintenance shall be limited to the scheduled (preventative) maintenance permitted by table II and unscheduled (corrective) maintenance as permitted by the reliability requirements of 3.6. The sets shall be capable of operating for 3000 hours without critical failures.

3.3.2 Operating speeds. Operating (rated) speed of mode II sets shall be 3,428 revolutions per minute (rpm). Operating speed (rated) of mode III and mode IV sets shall be 3,600 rpm. All sets shall be capable of operation at 4,000 rpm for a period of five minutes without damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.3.3 Rough handling. The set shall not be damaged (see 6.4.5) by rough handling which could be encountered during rail, truck, trailer, aircraft, or helicopter transportation (see 6.4.6, 6.4.7 and 6.4.8).

3.3.4 Storage temperature and humidity. The set shall not be damaged (see 6.4.2) by storage in temperatures from +160 °F (71.1 °C) to -65 °F (-53.9 °C) at any possible relative humidity within that temperature range and shall not be damaged by exposure to five consecutive 48 hour humidity cycles as specified in MIL-STD-705, method 711.1.

3.3.5 Lifting and towing provisions. The lifting attachment shall be capable of withstanding not less than eight times the weight of the set. The set shall remain within 15 degrees of level when lifted by the lifting attachment. The towing provisions shall be capable of withstanding a pull of not less than five times the total weight of the set.

3.3.6 Flexural vibration and critical speeds. The set shall be free from dangerous flexural vibration and dangerous torsional critical speeds (see 6.4.3 and 6.4.4) between 2,800 and 4,000 rpm.

3.4 Maximum power. The maximum output power (corrected) in kW for the first article sets shall be not less than 125 percent of rated load. The minimum acceptable maximum power (corrected) in kW for production sets shall be 95 percent of the average maximum power (corrected) of first article sets taken at the beginning of the reliability/endurance test).

3.5 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UM04 limits of class C2 equipment of MIL-STD-461.

3.6 Reliability/maintainability. The set shall have a specified meantime-between-failure (MTBF) of 500 hours and a lower test MTBF of 250 hours as defined in MIL-STD-781. The maintenance ratio (see 6.4.20) shall not be more than 0.04.

3.7 Engine. The engine, including all systems, components, accessories, and auxiliaries, shall conform to the applicable drawing and shall meet the requirements as specified herein.

3.7.1 Oil temperature. The oil temperature measured at the bottom of the oil dipstick shall stabilize between 100 °F (37.8 °C) and 275 °F (135 °C) at ambients between -25 °F (-31.7 °C) and 107 °F (41.7 °C). Above ambients of 107 °F (41.7 °C), the oil temperature shall not exceed 285 °F (140.6 °C).

3.7.2 Fuels. The set, when operating on DF-1, DF-2 or DF-A diesel fuel conforming to VV-F-800, or referee grade diesel fuel conforming to MIL-F-46162, shall meet the requirements specified herein at all loads up to and including rated load. All components of the set and fuel system shall be capable of continuous operation for a minimum of 400 hours on JP-4 turbine fuel conforming to MIL-T-5624 and with a maximum cetane rating of 45. When operating on JP-4 turbine fuel, the set shall meet all requirements herein except as follows:

a. The set shall be capable of starting and operating at ambient temperature from 0 °F (-17.8 °C) to 100 °F (37.8 °C).
b. Rated load may not be reduced more than 15 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

The set shall also be capable of starting and operating at all loads, up to and including rated load on JP-5 conforming to MIL-G-5624 and DF-M conforming to MIL-G-16884.

3.7.3 Governing system performance. The governing system, when properly adjusted, shall perform as specified herein.

3.7.3.1 Frequency (speed) regulation. The governing system shall maintain the frequency (see 6.4.9) within three percent rated frequency (speed) for all loads from no load up to and including rated load.

3.7.3.2 Short-term steady-state stability (30 seconds). With the regulation set at a maximum of 3.0 percent at every constant load from no-load to rated load, the system shall maintain frequency (speed) within a bandwidth equal to two percent of rated frequency (speed). The system shall not permit repetitive frequency (speed) variations, commonly called "hunting", even though within the allowable two percent band.

3.7.3.3 Long-tern steady-state stability (four hours). With the regulation set at a maximum of 3.0 percent at constant ambient temperature, constant barometric pressure, constant voltage and any constant load from no-load to rated load, the system shall maintain frequency (speed) within a bandwidth equal to three percent of rated frequency (speed) in a four hour operation period.

3.7.3.4 Transient performance. Following any sudden increase in load, including from no load to rated load, the governing system shall re-establish stable engine operating conditions (see 6.4.10) within four seconds, and the maximum transient frequency (speed) change below the new steady-state frequency (undershoot, see 6.4.11) shall be not more than four percent of rated frequency (speed). Following any sudden decrease in load, including from rated load to no load, the governing system shall re-establish stable engine operating conditions within six seconds, and the maximum transient frequency (speed) change above the new steady-state frequency (overshoot) shall be not more than five percent of rated frequency (speed).

3.7.3.5 Frequency (speed) adjustment range. The manual speed control shall permit adjustment of plus or minus three percent of rated frequency (speed) at all load conditions from no load to rated load.

3.7.4 Engine protective device, low fuel shutdown. The set shall be equipped with a low fuel shutdown protective device that shall shut down the set when the fuel level is between 1/2 inch and 1-1/2 inches above fuel pickup.

3.8 Generator.

3.8.1 Generator rotor assembly. The rotor assembly shall be in electrical and mechanical balance at all speeds up to 125 percent rated speed. The rotor shall withstand operation at 125 percent rated speed for 15 minutes without damage.

3.8.2 Winding resistance, stator, and field winding. The winding resistance of each winding shall not differ by more than ±10 percent from the resistance specified on the applicable drawing. All resistance values shall be corrected to a temperature of 77 °F (25 °C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.8.3 Insulation resistance. The insulation resistance of all circuits shall be not less than one megohm when the circuits are stabilized at an ambient temperature of 77 °F (25 °C) ±9 °F (5 °C).

3.8.4 Temperature rise. The maximum temperature rise, with the set delivering rated load continuously, shall be not more than 135 °F (75 °C) for the stator windings and connections when measured by the rise in resistance method.

3.8.5 Dielectric strength. Electric windings shall withstand the following 60Hz voltage applied for one minute:

a. Generator armature: 1,480V between phase windings, and 1,240V between phase windings and ground.
b. Generator field and exciter windings: Ten times ceiling voltage but not less than 1,500V nor more than 3,500V applied between windings and ground.
c. Windings energized by the 24V dc control, cranking and battery charging system: 500V applied between the winding and ground.
d. All others: Twice rated voltage plus 1,000V applied between the windings and ground or between windings, as applicable.

3.8.6 Short circuit. The generator and control box assembly operating as a unit, shall withstand 10 consecutive 10-second line-to-line or line-to-neutral short circuits applied at 60-second intervals at the generator output terminals when operating at rated load and frequency (speed) without reduction of dielectric strength to a point where it will not meet the requirements of 3.8.5. Sets shall be operating at rated load when the short circuits are applied.

3.8.7 Voltage transient performance. Performance of the set under transient conditions shall not exceed that specified in the detailed specification (see 3.1).

3.8.8 Voltage waveform. The deviation factor of the line-to-neutral and line-to-line voltages for three phase generator connections at rated load at 0.8 pf; rated load at unity pf; and no-load shall not exceed 5.0 percent. Under the same conditions, the deviation factor at the single phase connections shall not exceed 6.0 percent. Single frequency harmonics shall not exceed 3.0 percent. Voltage waveform is not applicable for mode IV sets.

3.8.9 Voltage unbalance. Maximum variation between line-to-line voltages shall not exceed five percent of rated voltage for the three phase, 120/208 volt connection under the conditions of a single phase, line-to-line, unity pf, load of 25 percent of rated current and no other load on the generator. Voltage unbalance is not applicable for mode IV sets.

3.8.10 Phase balance (voltage). With the generator connected for three phase, 120/208 volt output, the maximum difference in the line-to-line and line-to-neutral voltages under open circuit at rated voltage and speed shall not exceed one percent of the rated line-to-line or line-to-neutral voltage. Phase balance is not applicable for mode IV sets.

3.8.11 Phase sequence (rotation). Phase rotation for AC sets shall be L_1, L_2 and L_3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.8.12 Voltage performance. The generator, together with the regulation systems, shall automatically keep the AC voltage within the limits specified herein.

3.8.12.1 Voltage regulation. The voltage regulation no load to rated load and from rated load to no load shall be not more than three percent of the rated voltage for all voltage connections throughout the voltage ranges specified in 3.8.12.5.

3.8.12.2 Voltage short-term stability (30 seconds). At any constant load from no load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to two percent of rated voltage.

3.8.12.3 Voltage long-term stability (four hours). At constant ambient temperature, constant barometric pressure, constant frequency (speed) and any constant load from no load to rated load, the voltage shall remain within a bandwidth of two percent of rated voltage in a four hour operating period.

3.8.12.4 Voltage dip and rise for rated load. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows: With the set initially operating at rated frequency and rated voltage and following any sudden change in load from no load to rated load, the instantaneous terminal voltage shall not dip more than 30 percent of rated voltage and shall read stable conditions (see 6.4.10) within three seconds. Overshoot or undershoot of the final voltage shall not exceed the initial voltage transient in amplitude. The above requirements shall also apply when the load is suddenly changed from rated load to no load, except that the initial voltage transient shall involve a voltage rise not to exceed 30 percent of rated voltage.

3.8.12.5 Voltage adjustment range. The minimum voltage adjustment range for all voltage connections shall be five percent above and below rated voltage. This voltage adjustment range shall apply for all loads between no load and rated load under all temperature conditions for operation as specified in 3.3. Clockwise rotation of voltage adjustment control shall increase set output voltage.

3.8.13 Field currents. The generator exciter field current of production generators shall not exceed the values specified in the detailed specification (see 3.1).

3.8.14 Generator efficiency. The efficiency (see 6.4.16) of the generator and excitation system at normal ambient temperature shall be as specified in the detailed specification (see 3.1).

3.9 Circuit breaker. The circuit breaker shall open within 0.05 seconds under the condition of a short circuit in any one phase or in all phases simultaneously.

3.10 Control system. All dc control devices shall be suitable for operation on voltages of 20 to 32 volts. Total dc current for all control devices shall not exceed five amperes when the set is operating. The control system circuit breaker and reverse battery polarity device shall prevent damage to the control circuits and alternator in the event of reversed battery polarity. When the set is not operating, and with the master switch in the "OFF" position, battery

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

current drain shall not exceed five milliamperes.

3.11 Instruments. The allowable error (see 6.4.12) of each metering system, including the electrical circuit, under all conditions specified in 3.3.1, shall be as specified herein.

3.11.1 Voltmeter system. The accuracy of the voltmeter system shall be within three percent.

3.11.2 Ammeter system. The accuracy of the ammeter system shall be within 10 percent.

3.11.3 Frequency meter system. The accuracy of the frequency meter system shall be within 1.5Hz for 60Hz operation and 6Hz for 400Hz operation.

3.12 Delivery connection tag. Unless otherwise specified (see 6.2), each set shall be connected for 120/208 volts, three phase, and shall have a connection tag attached to one of the load terminals indicating: "This set is connected for 120/208 volts. See connection changeover instructions for other voltage ratings". Delivery connection tag is not applicable for mode IV sets.

3.13 Audio noise. Audio noise sound pressure levels emanating from the set when it is operating at any load up to and including rated load shall not exceed 80 dBA when the microphone is located 1.2 meters above the ground, seven meters measured in any direction from the perimeter of the set. When specified (see 6.2), an optional noise control kit, suitable for use with any mode, shall be provided to limit the audio noise sound pressure emanating from the sets to 70 dBA or less at seven meters and 85 dBA maximum at one meter from the control panel under all operating conditions specified herein.

3.14 Ground rod assembly. When specified (see 6.2), a ground rod assembly in accordance with W-R-550, shall be supplied.

3.15 Treatment and painting. The set shall be treated and painted in accordance with the applicable drawings (see 3.1). The following items shall not be painted: terminal wiring connections, governor linkage, instruction diagrams, nameplates, rectifiers, relays, circuit breakers, switches, and all other parts whose operation or function would be adversely affected by paint. After assembly and testing, any chips or scratches in painted surfaces shall be touched up, in accordance with MIL-C-53072, prior to shipment using the specified primer and/or finish paint.

3.16 Workmanship, safety, and human engineering. Workmanship, safety, and human engineering shall be in accordance with MIL-STD-454, requirements 9, 1, and 62, respectively. In addition, workmanship shall be of a quality to assure delivery of sets which are free from defects resulting from defective material and improper manufacturing or assembly practices. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to repeat any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Components and material inspection. The contractor is responsible for insuring that components and material used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3. Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.18 and 6.6).

4.1.4 Disassembly inspection. Failure of any examination or test by the first articles shall be cause for disassembly, in the presence of a Government representative, of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.5 Inspection failure. Should the Government elect to perform or repeat any inspection or test in this specification, failure of a set to meet any requirement specified herein shall be cause for refusal by the Government to accept production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on any sets previously produced under the contract. Any deficiencies found as the result of such elective inspection or test will be considered prima facie evidence that all sets accepted prior to the completion of such inspection or test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. The provisions of this paragraph apply, notwithstanding any prior acceptance of first articles or first article test reports.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Individual generator inspection (see 4.4.1).
c. Individual set inspection (see 4.4.2).
d. Qualify conformance inspection (see 4.4).
e. Inspection comparison (see 4.7).
f. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the generator or set. Evidence that any components or subassemblies do not comply with the requirements of the drawings shall be cause for rejection of that component or subassembly.

4.3.2 Examination of generator and sets. Examination of the generator (with exciter-regulator system) and the sets shall be made without disassembly. Evidence that the generator (with exciter-regulator system) or sets do not comply with the drawings shall be cause for rejection of the generator or sets.

4.3.3 Tests. The generator with exciter-regulator system and the sets shall be subjected to the tests marked "X" in column 1 of table I. The set tests marked "X" under column 1 of table I may be conducted in any order, except that following the "Insulation Resistance Test", the first three set tests shall be "Voltage Waveform", "Railroad Impact", and "Humidity" in that order, and the last test shall be "endurance run". The reliability test shall be performed on the two sets furnished specifically for the reliability test, when applicable. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.3.

4.3.4 Inspection report. The contractor shall furnish, within the time frame specified (see 6.2), an inspection report in accordance with MIL-STD-831 covering first article inspection. The inspection report shall also include the following:

a. All test data obtained from applicable tests.
b. Description of malfunctions, damages, failures, and adjustments (other than adjustments permitted by this specification) which occur during inspection.
c. Causes and analysis of malfunctions, damages or failures and reasons for adjustments.
d. Corrective action taken or required.

4.4 Quality conformance inspection.

4.4.1 Individual generator inspection. Inspection shall be performed prior to assembly into a set.

4.4.1.1 Examination. Each generator (with exciter-regulator system) shall be examined without disassembly for conformance to the drawings. Evidence that the generator (with exciter-regulator system) does not conform to the drawings shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.4.1.2 Tests. Before final assembly of the generator, each generator shall be subjected to the first two tests marked "X" under column 3 of table I. Failure of any test shall be cause for rejection.

4.4.2 Individual set inspection.

4.4.2.1 Examination. Each set shall be examined without disassembly for conformance to the drawings. Evidence that the set does not conform to drawings shall be cause for rejection.

4.4.2.2 Tests. Each set shall be subjected to the tests marked "X" under column 3 of table I with new air, fuel, and oil filters installed. Failure of any test shall be cause for rejection.

4.4.3 Lots.

4.4.3.1 Generator with regulation system assemblies. Lots shall be formed from generators which have successfully passed the inspection specified in 4.4 and regulation system assemblies that have successfully passed the inspection specified in 4.4.1 and 4.4.2.

4.4.3.2 Sets. Lots shall be formed from generator sets which have successfully passed the inspection specified in 4.4.2. The first lot shall be not less than 25 sets if more than 25 are ordered.

4.4.4 Sampling for generator with regulation system assemblies and sets. Sampling size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.5 Tests. Samples selected in accordance with 4.4.4 shall be subjected to the tests marked "X" in column 4 of table I. No adjustment or substitution of components shall be made on the sets selected for sample inspection. These tests shall be performed utilizing the auxiliary fuel provision.

4.5 Test schedules. The test schedule shall be as shown in table I. Nonconformance to the applicable requirement when tested as shown in column 6 of table I shall constitute faulure of these tests. The Government reserves the right to reject the equipment for not meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in section 3 and table I shall apply to all tests performed as a part of another test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE I. Test schedule.

					Test (see Note 4)	Test Method (MIL-STD-705) and/or Test Paragraph (see note 1)
	1	2	3	4	5	6
1	X	-	X	-	High potential (generator only)	302.1, test each winding before installation of rectifiers, see note 2.
2	X	-	X	-	Winding resistance measurement (generator only)	401.1, measure all windings before installation of rectifiers using kelvin or wheatstone bridge methods.
3	X	-	-	X	Insulation resistance, stator only	301.1, 4.3.3
4	X	-	-	X	Overspeed (generator)	505.3
5	X	-	-	X	Generator power input	415.1 (also at 120/208 V, 3 ph for mode II and III).
6	X	-	-	X	Open circuit saturation curve	410.1 (one point at rated voltage).
7	X	-	-	X	Rated load saturation curve	413.1 (one point at rated voltage).
8	X	-	-	X	Short circuit (mechanical strength)	625.1
9	-	X	-	-	Railroad impact	740.5, 4.6.6
10	-	X	-	-	Drop (freefall)	740.2, 4.6.8
11	-	X	-	-	Drop (ends)	740.3, 4.6.8
12	-	X	X	-	Frequency (speed) and voltage regulation, stability and transient response (short term)	608.1
13	-	X	-	X	High temperature at 125 °F (51.7 °C)	710.1
14	-	X	X	-	Overspeed (set)	505.1
15	-	X	X	-	Instrument check	513.1
16	-	X	X	-	Phase sequence (rotation)	507.1 (at 120/208 V, 3 ph and 120 V, 3 ph only: mode II and III sets only).
17	-	X	X	-	Frequency (speed) adjustment range	511.2, at no load and load.
18	-	X	X	-	Ripple voltage	650.1 (mode IV sets only).
19	-	X	X	-	Start and stop	503.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE I. Test schedule. (cont'd)

	1	2	3	4	Test (see Note 4)	Test Method (MIL-STD-705) and/or Test Paragraph (see note 1)
	1	2	3	4	5	6
20	-	X	X	-	Reverse battery polarity	516.5
21	-	X	-	X	Phase balance (voltage)	508.1 (at 120/208 V, 3 ph only, mode II and mode III sets only).
22	-	X	-	-	Audio noise	661.2, 3.13, 4.6.3
23	-	X	-	X	Voltage unbalance with unbalanced load	620.2 (120/208 V, 3 ph only mode II and III sets only).
24	-	X	-	X	Fuel level switch	515.5, 3.7.4
25	-	X	-	X	Voltage waveform (harmonic analysis)	601.4 (mode II and mode III sets only), see note 3.
26	-	X	-	X	Voltage dip and rise for rated load	619.2
27	-	X	-	X	Endurance run	690.1 (for 200 hours)
28	-	X	-	-	Long term frequency (speed) and voltage stability	608.2
29	-	X	-	-	Humidity	711.1 (measure insulation resistance stator windings only).
30	-	X	-	-	Lifting and towing	740.4
31	-	X	-	-	Maximum power	640.1
32	-	X	-	-	JP-4 fuel test	3.7.2
33	-	X	-	-	JP-5/DF-M fuel test	3.7.2
34	-	-	X	-	Maximum power	640.4
35	-	X	-	-	Reliability/endurance	695.1, 690.1 (see 4.6.4) Test method 720.1 shall be performed before and after reliability test. Altitudes shall be 8000 feet (564.9 mmHg), 95 °F

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE I. Test schedule. (Cont'd)

					Test (see Note 4)	Test Method (MIL-STD-705) and/or Test Paragraph (see note 1)
	1	2	3	4	5	6
						(35 °C) and 5000 feet (632.4 mmHg) 107 °F (41.7 °C). One set shall run 200 hours using JP4 fuel).
36	-	X	-	-	Road test	4.6.7
37	-	X	-	-	Torsiographing	504.2, 3.3.6
38	-	X	-	-	Rain	711.3 (at 4 inches per hour)
39	-	X	-	-	Safety and Human Engineering	Government approval, 3.17
40	-	X	-	-	Altitude Operation	720.1 (at 8000 ft (564.9 mmHg) 95 °F (35 °C). Take temperature connections), 3.3.1.1.1 rise measurements on all stator windings), 3.3.1.2
41	-	X	-	-	Starting and operating (Moderate Cold Battery Start)	701.2 (at -25 °F) (31.7 °C) (511.1 at all voltage
42	-	X	-	-	Vibration Analysis	4.6.5
43	-	X	-	-	Inclinded Operation	660.1, 3.3.1.1.1
44	-	X	-	-	Electromagnetic interference	3.5, 4.6.2
45	-	X	-	-	Starting and Operating (Extreme) Cold Battery Start)	701.1 at -50 °F (-45.6 °C) 3.3.1.1.1
46	-	X	-	-	Voltage waveform (Oscillographic)	601.1 at no load, rated load at 0.8 pf, and rated load, 1.0 pf at all voltage connections.
47	-	X	-	-	Voltage Modulation	602.1 at no load, rated load at 0.8 pf, and rated load at 1.0 pf, at all voltage connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE II. Servicing, adjustment and replacement allowed for all testing.

Item No.	Minimum Interval (Hrs of Operation)	Servicing, Adjustment and Replacement Allowed (Scheduled Maintenance)
1	12	Inspect set for loose connections, leaks, cracks, make minor adjustments, if necessary. Record all maintenance actions.
2	12	Check lubricating oil. Add oil if necessary.
3	12	Drain contaminants from fuel filters and water separator through drain valves.
4	12	Check air cleaner restriction indicator. Service filter as necessary.
5	12	Check battery. Service as necessary.
6	125	Change lubricating oil and filter.
7	125	Change all fuel filter elements. Clean strainer.
8	125	Lubricate linkage and clean cooling fins.
9	250	Check and clean injection nozzle if required.
10	500	Replace water separator.
11	1500	Replace injection nozzle.
12	3000	Overhaul engine.

NOTES:

1. All mode II and III tests performed in single phase, 120 V, two-wire configuration unless otherwise specified in table I. Additional test conditions, when required, are shown in Column 6.

2. High potential tests may be made after assembly of the engine to generator but before final assembly of complete set. On first article sets, perform high potential test on all set windings (see 3.8.5).

3. On first article sets, measure all line-to-line and line-to-neutral voltage harmonics at each voltage connection for no load, rated load and rated kW at 1.0 power factor. On quality conformance sets measure only the voltage harmonics at the condition which yielded the greatest harmonic during the first article set test.

4. The Government reserves the right to reject the set for not meeting any requirement of this specification even though not performing a test directly related to the specific requirement.

4.6 Test procedures.

4.6.1 Test condition. Tests shall be conducted in accordance with MIL-STD-705, MIL-HDBK-705, and as specified herein. Referee grade diesel fuel conforming to MIL-F-46162, grade I or II as applicable to the set environment, shall be used for all tests except the JP-4 test and the JP-5/DF-M fuel test. Lubricants specified below shall be used during all tests in accordance with the ambient temperature range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Specification	Grade	Daily Low Ambient Temperature (°F)
MIL-L-2104	30	above 32 °F
MIL-L-2104	10	-10 °F to +32 °F
MIL-L-46167	arctic	below -10 °F

Test instruments shall be of the laboratory type and shall have been calibrated within 30 days of the start of testing and at six month periods thereafter. Instruments used in calibration should have at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading instruments shall have at least 0.5 percent accuracy and shall be connected to indicate in the most accurate portion of their range. On dc instruments, the readings shall not be made on the lower 15 percent of the scale. On ac instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of voltage and frequency recording type meters, the Texas Instrument Company, Model PDRHXFHXVA-A16-XT or the Gould Part Number 2108-2202-005542 shall be used. Oscillograph galvanometer frequency response shall be flat (+5 percent) from zero to 3,000 Hz. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warm-up period of the set and record continuously for the duration of the test. The recording meters shall be operated at a minimum speed of 6 inches per hour during the portions of the test where steady state loading conditions exist and shall be operated at a minimum speed of six inches per minute at least 30 seconds, before, during and after a load change. Exhaust smoke conditions for sets during all maximum power and altitude testing shall be measured by the use of a Robert Bosch EFAW 65 sampling pump and analyzed on a EFAW 68 analyzing instrument at room temperature (65 °F (18.3 °C) to 80 °F (26.7 °C)). The smoke sampling tube shall be located in the exhaust pipe, positioned midway of and parallel to, the pipe walls and between 12 and 60 inches from the exhaust manifold. At that location, the exhaust pipe inside area shall not be larger than 125 percent of the exhaust manifold outlet area. Tubing between the sampling pump and exhaust probe shall not exceed 24 inches in length and shall have an inside diameter of not more than 1/4 inch.

4.6.2 Electromagnetic interference test. The electromagnetic interference test shall be performed in accordance with MIL-STD-462 and Appendix A. The test shall be conducted at all rated voltage connections at the following load connections:

a. No load.
b. Rated load at 1.0 pf.
c. Rated load at 0.8 pf.

4.6.3 Audio noise test. Measure audio noise sound pressure levels at no load and rated kW load. The microphone(s) shall be located 1.2 meters high above the ground at positions seven meters from the perimeter of the set at the center of each side and at each corner (eight positions spaced 45 degrees apart). Also measure audio noise sound pressure levels at the operator's position one meter out from the center of the control panel and at a height of 1.2 meters above the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

ground. Sound pressure levels shall be measured in accordance with MIL-STD-1474 for generator sets. The tests shall be conducted on sets (see 3.13) and on sets equipped with audio noise control kits (see 6.2).

4.6.4 Reliability/endurance test. The reliability/endurance test consists of a reliability test portion followed by an endurance test portion. The test shall be conducted on two sets of each mode which have completed and passed all first article tests excluding the altitude test, which shall follow the reliability/ endurance test. In each test portion, the servicing, adjustment, and maintenance of all parts, assemblies and accessories shall be in accordance with table II. No maintenance other than that allowed by the maintenance schedule shall be allowed after the reliability test portion and prior to the endurance test portion.

4.6.4.1 Reliability test. The reliability test will be performed as specified in MIL-STD-705, test method 695.1. Two sets of each mode shall operate for at least 2350 hours and the total test duration for both sets shall be 5000 hours. The upper test mean time between failure is 500 hours, the lower test mean time between failures is 250 hours, as defined in MIL-STD-781. Accept the sets of a mode if the total number of relevant, chargeable failures is 15 or fewer. Reject the mode if the total number of relevant, chargeable failures is 16 or more, or at the government option, at any time that the total test hours divided by the number of failures is less than 150. For the accept/reject decision, failures are defined in section 6.4.14 and MIL-STD-781.

4.6.4.2 Endurance test. The endurance test will be performed as specified in MIL-STD-705, test method 690.1. After successfully completing the reliability portion or the test, continue to run both sets of each mode until each set has operated 2980 hours. Accept the mode if no critical failures occur during the reliability or the endurance test portions on both sets of the mode. Reject if one or more critical failures occur. Critical failures are defined in 6.4.14.3.

4.6.4.3 Maintainability. The sets shall be evaluated for compliance with the maintainability requirements of 3.3.1.2 and 3.6 during the reliability/endurance test. All maintenance data will be recorded to include: time of occurrence of all scheduled and unscheduled maintenance, what action was performed, maintenance time, and the parts and supplies required. Accept if the maintenance ratio is 0.04 or less. Reject if the maintenance ratio is greater than 0.04. The maintenance ratio is defined in 6.4.17. During the reliability/endurance test, when an incident occurs, repairs shall be limited to fixing only the cause of the failure. Incipient problems which must be repaired or adjustments which are needed shall be performed in separate maintenance actions.

4.6.5 Vibration analysis. The indexing (orientation) of the generator to the engine shall be random in nature. The generators shall not be indexed to the engines based on vibration so as to minimize vibration of the set. On the first article sets, accelermoters shall be located on the outside of the components specified below so as to best measure their vibration in the horizontal, vertical and transverse planes:

a. Front engine main bearing.
b. Rear engine bearing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

c. Rear generator bearing.
d. Control panel in center of area determined to have the maximum amount of vibration.

4.6.5.1 Procedure. The set shall be placed on level floor ±0.25 inches maximum difference in elevation under set. The total weight of the set shall be supported on vibration isolator pads located under the four corners of the set. The pads (Isomode pad, VIBX pad or equal) shall have a short diameter of 45, a thickness of 5/16 inch unloaded and have an area exposed to the load of the set so as to result in a pressure on the pad of 50 ±5 psi. The set fuel tank shall be over 90 percent full. The location selected for the accelermoters shall be accurately documented on all first article sets and shall be located within 0.125 inch of the documented location. The set shall be run at rated speed and load and at no load and the broad band acceleration (in peak G's) recorded for each load, location end plane specified. In addition, a sweep from 5 to 1,000 Hz shall be recorded on a card no smaller than 4 inches by 6 inches by an automatic sweep plotter for each load, location and plan specified above. Once a recorder plotter is selected, all measurements shall be on the same make and model recorder.

4.6.5.2 Sample test. Vibration data shall be obtained by the same procedure as in 4.6.5.1 but at rated load only.

4.6.6 Railroad impact. Perform method 740.5 of MIL-STD-705 with the sets positioned 180 degrees with respect to each other for a total of four impacts at eight miles per hour (±0.5 mph). Two impacts shall be made with sets positioned longitudinally with respect to the axis of the car and two impacts with the set rotated 90 degrees from the longitudinal position. The sets shall not be packaged for the railroad impact tests.

4.6.7 Road test. The road test consist of four 140-mile cycles as defined in 6.4.7. Perform method 608.1 of MIL-STD-705 before start and after the completion of entire road endurance run. At the conclusion of each 140-mile cycle, each generator set shall be visually examined for any damage and operationally checked at no load.

4.6.8 Drop tests. The drop tests shall be performed in accordance with MIL-STD-705, methods 740.2 and 740.3. Test method 608.1 shall be performed at rated load only, both before and after the drops. For method 740.2, the height shall be 12 inches and position of drop shall be flat. For method 740.3, the height of the drop shall be six inches.

4.7 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to the examination and test specified in section 4 as First Article Tests or Quality Conformance Tests to determine that the quality of the selected sets is equal to the quality standards established during first article testing. The inspection will be performed by the Government at a site selected by the Government. Sets will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article sets.

4.7.1 Inspection failure. Failure of a set to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the sets and shall be cause for refusal by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection will be considered prima facie evidence that all sets previously accepted are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-G-28554.

5. PACKAGING

5.1 Preservation, packing, and marking. Packaging requirements shall be level A, B, or C as specified (see 6.2), and shall be in accordance with MIL-G-28554. All mode II and III sets shall be shipped connected for 120/208 volts. New replaceable fuel, oil and air filters shall be installed.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The generator sets specified herein are intended to supply power for multipurpose use in military applications.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification, and of the applicable specification sheet (see 2.1.1).
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of the first articles and number to be furnished (see 3.2).
d. Whether two additional first article model sets for performing the reliability test are required (see 3.2).
e. Whether ground rods shall be provided with each set (see 3.14).
f. Time frame for submission of the first article inspection report (see 4.3.4).
g. Whether sets are to be shipped connected for voltage other than as specified (see 3.12).
h. Level of preservation and packing required (see 5.1).
i. Whether all system tests shall be performed other than as specified (see 4.7.1).
j. Whether sets are to be furnished with audio noise control kits per TA 88-13560 (see 3.13).
k. Whether sets are to be furnished with winterization kits per TA 84-13108 (see 3.3.1.1.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Start. A set is considered to have started when it is operating at rated voltage and speed without the further use of starting aids.

6.4.2 Temperature and humidity damage. Temperature and humidity damage is defined as conditions causing malfunction of any component or part, corrosion, breakage, deformation, or reduction of insulation resistance below 50,000 ohms.

6.4.3 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibration exceeds 9,000 psi.

6.4.4 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibration stress in the shaft from torsional vibration exceeds 5,000 psi.

6.4.5 Rough handling damage. Rough handling damage is defined as any condition resulting in malfunctioning of the set, deformation, loosening, breakage, or change of fit of any component or part.

6.4.6 Railroad transportation. Railroad transportation shall be interpreted to mean humping speed up to and including eight miles per hour under the test conditions specified in MIL-STD-705, method 740.5.

6.4.7 Truck or trailer transportation. Truck or trailer transportation is defined as the conditions encountered during four cycles of a road endurance test, each cycle consisting of the following, with the sets mounted on the specified trailer:

Road Conditions	Distance	Speed (MPH)
a. Paved highway	50 miles	up to 50
b. Level cross country	50 miles	up to 20
c. Hilly cross country	25 miles	up to 20
d. Belgian block	15 miles	up to 20

6.4.8 Aircraft and helicopter transportation. Aircraft and helicopter transportation shall be interpreted to mean a 12 inch free fall drop and the six inch end drop under the test conditions specified in MIL-STD-705, methods 740.2 and 740.3.

6.4.9 Frequency (speed) and voltage regulation. Frequency (speed) and voltage regulation is defined as the difference between the no load value and the rated load value divided by the rated load value. To express regulation as a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

percentage, multiply the value found by 100.

6.4.10 Stable set operating conditions (voltage and frequency). Stable set operating conditions are defined as the conditions specified for steady-state performance.

6.4.11 Overshoot and undershoot. Refer to MIL-STD-705, figure 608.1-IV for definitions of overshoot and undershoot.

6.4.12 Meter accuracy/error. Accuracy is a number which defines the limit of error expressed as a percentage of full-scale value. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indications, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.4.13 Ambient conditions. Ambient conditions are considered to be at sea level with the ambient temperature between 60 °F (15.6 °C) and 85 °F (29.4 °C).

6.4.14 Failure. A failure is defined as the inability of an item to perform within specified limits. Relevant, chargeable failures as defined in MIL-STD-781 shall be used to evaluate compliance with the reliability requirement and testing of 3.6 and 4.6.4. Repetitive failures will not be allowed. The contracting officer will identify all failures and will classify them into the following categories.

6.4.14.1 Relevant failure. Any failure of an item which prevents the set from meeting the requirements specified in section 3.

6.4.14.1.1 Failure detection. A failure which would prevent the set from meeting any requirement specified herein is relevant notwithstanding the fact that it is detected during a shutdown for service and did not cause the set to perform out of limits during the previous testing (e.g., a shutter failure on a 40 °F (4.4 °C) day might not cause malfunction, but would preclude operation at extreme temperatures and is therefore relevant, even if detected during shutdown for service).

6.4.14.2 Nonrelevant failure. Any failure of an item which is not used to compute reliability. Examples of nonrelevant failures are as follows:

a. Secondary failures caused by failures in the powered equipment or other occurrences in the set environment when integral set protection is not provided against such equipment failure or occurrence, e.g., explosion or fire.
b. Failures which may be corrected by normal operator function, e.g., readjustment of voltage after the four hour long term stability period.
c. Failures due to characteristics of the load, e.g., waveform distortion caused by saturated inductors.
d. Failures resulting from operating items beyond requirements, e.g., if belts are run to failure to determine mean life, failures after 1,000 hours are not failures.
e. A failure of an item which does not prevent the set from meeting the power output requirements specified herein, e.g., a panel light burns out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

f. Failure due to operator error when proper procedures are documented in technical manuals, instruction plates mounted on the set, or both; e.g., use of improper lubricant.
g. Failures due to proper performance of erroneous and/or inadequate operation and maintenance instructions.

6.4.14.3 Critical failure. A critical failure is defined as a relevant failure requiring removal of cylinder head, oil pan or gear cover to repair. Critical failures are used to determine compliance with the requirement of 3.6.

6.4.15 Fuel and oil resistant. Fuel and oil resistant is defined as capable of a one second immersion in any petroleum oil lubricants used in the sets, every 24 hours for a total of 240 hours without degradation to the point where it will no longer serve its intended use, or in the case of engine generator mounts, to the point where they will not pass the railroad impact test.

6.4.16 Generator efficiency. Generator efficiency is defined as output load multiplied by 100, divided by generator input power required for that load.

6.4.17 Maintenance ratio. The maintenance ratio (MR) is the ratio of the number of manhours required for all scheduled and unscheduled maintenance per hour of set operation. Manhours at the operator/crew, organization, and direct support levels are included but not manhours to repair components or perform daily operational checks.

6.4.18 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.4.19 Rated load. Rated load is rated kW at rated power factor, rated frequency (speed) and rated voltage.

6.5 First article sets. Any changes or deviations of production sets from the approved first article sets (preproduction model) during production will be subject to the approval of the contracting officer. Approval of the first article (preproduction model) will not relieve the contractor of his obligation to furnish sets conforming to this specification.

6.6 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Low temperature. The sets described by this specification are suitable for storage at temperatures to minus 65 °F (minus 53.6 °C).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

6.8 Configuration management. This specification is the configuration management base line for the sets described herein. Configuration management procedures are in accordance with DIAR 4120.7.

6.9 Subject term (key word) listing.

Generator sets
Generator sets, DED
Tactical utility

6.10 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: H. E. Taylor 8/14/89).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6115-A520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

APPENDIX A

ELECTROMAGNETIC EMISSION AND SUSCEPTIBILITY REQUIREMENTS FOR THE CONTROL OF ELECTROMAGNETIC INTERFERENCE

10. SCOPE

10.1 Scope. This document clarifies the limits, test methods and general procedures to the used for measurement of EMI on Mobile Electric Power Generating Sources (MEPGS). Such clarification is necessary as a result of the recent change to MIL-STD-461. Clarification as to the use of MIL-STD-462 is also included. This document is intended to be used as a clarifying supplement to MIL-STD-461 and MIL-STD-462 and does not replace MIL-STD-461 or MIL-STD-462.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. GENERAL REQUIREMENTS

30.1 Class C equipment requirement. MIL-STd-461 established Class C2 for engine generators and associated components, Uninterruptible Power Sets (UPS) and Mobile Electric Power (MEP) equipment supplying power to or used in critical areas. The UMO4 limits for conducted emissions, radiated emissions, and radiated emissions susceptibility listed in Part 9 of MIL-STD-461 is applicable to sets rated at 240 kVA or less. The UMO4 limits for radiated emissions and radiated susceptibility may be extended to sets with a rating greater than 240 kVA when specified in the procurement document.

30.2 Limits and frequency ranges. Part 9 of MIL-STD-461 establishes the following information:

(1) Conducted Emissions, 0.015 to 50 MHz, Power Leads.
(2) Radiated Emissions, 14 kHz to GHz, Electric Field.
(3) Radiated Susceptibility, 2 MHz to 10 GHz, Electrical Field.

Figures 9-1 and 9-2 of Part 9 establish the criteria for broadband conducted and radiated emissions. The criteria for radiated susceptibility is that the set shall not exhibit any malfunction, degradation of performance, or deviation from specified indications beyond the tolerances indicated in the individual equipment specification when subjected to the following radiated E-fields in the frequency range of 2 MHz to 10 GHz:

Frequency Range	E-Field (Volts/Meter)
2 to 400 MHz	10
400 MHz to 10 GHz	5

Above 30 MHz the requirement shall be met for both horizontally and vertically polarized waves.

40 QUALITY CONFORMANCE INSPECTION

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

40.1 Test procedures. Applicable test procedures for determining conformance to UM04 limits are contained in Notice 3 (EL) to MIL-STD-462. Specifically, Methods CE04, RE02, and RS03 shall be used for Conducted Emissions, Rated Emissions, and Radiated Susceptibility respectively. These test procedures are clarified by the attached supplementary test methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 1

CONDUCTED EMISSIONS

1. Test method. MIL-STD-462, Notice 3 (EL), Method CE04, Procedure (A) Conducted Emissions 50 kHz to 50 MHz, Power Leads.

2. Frequency limits. 0.015 to 50 MHz.

3. Applicability. Applicable in testing to requirements of figure 9.1 of Part 9 of MIL-STD-461.

4. Apparatus.

a. Delete: The use of Line Impedance Stabilization Networks (LISNs) and 50 OHM resistive terminations is not required.
b. Add: Fifty foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure. With the 50-foot cable connected to the MEPGS output and energized, determine the broadband conducted emissions (0.015 - 50 MHz) on each conductor at the end of the cable at no load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Conducted emissions exceeding the limit of figure 9.1 of MIL-STD-461 shall be cause for rejection.

7. Other.

a. The 50-foot cable shall be energized during the no load test.
b. The 10 microfarad feedthrough capacitors used in superseded test methods are not used in this test method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 2

RADIATED EMISSIONS

1. Test method. Notice 3 (EL), Method REO2, Radiated Emissions, Electrical Field.

2. Frequency limits. 14 kHz to 1 GHz.

3. Applicability. Applicable in testing to requirements of figure 9.2 of Part 9 of MIL-STD-461.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure.

a. In lieu of a continuously welded bond for the counterpoise, bonding straps shall be used.
b. With the 50-foot cable connected to the MEPGS output and energized, determine the radiated emissions (14 kHz - 1 GHz) at no load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Radiated emissions exceeding the limit of figure 9.2 of MIL-STD-461 shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no load test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 3

RADIATED SUSCEPTIBILITY

1. Test method. Notice 3 (EL), Method RS03, Radiated susceptibility, 14 kHz to 12.4 GHz, Electrical Field.

2. Frequency limits. 2 MHz to 10 GHz.

3. Applicability. Applicable to testing MEPGS.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The ratings of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor. The output monitor shall be a recording voltage and frequency meter IAW MIL-HDBK-705.

5. Procedure.

a. With the 50-foot cable connected to the MEPGS and energized, subject the set to electric fields of 10 V/m (2-400 MHz) and 5 V/m (400 MHz to 10 GHz) at no load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.
b. The set output voltage shall be monitored throughout the test with the recording voltage and frequency meter operating at six inches per hour until voltage and frequency variations as a function of field strength or frequency are noted. Upon observing such variation, the chart speed shall be increased to six inches per minute and the frequency spectrum in question shall be rescanned for acceptance/rejection analysis.

6. Failure criteria. Failure of the output voltage or frequency to remain within the 30 second short-term bandwidth (specified in the procurement) when subjected to the specified electrical fields shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no load test.

☆U.S. GOVERNMENT PRINTING OFFICE: 1989 - 604-033/41116

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
USA Belvoir, RDE Center
STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-53082(ME)	Generator Sets, Diesel Engine Driven 3kW, 60 Hertz, 400 Hertz, and 28 Volt Direct Current

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 22 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.